CASE STUDIES IN DATA SCIENCE · INDIVIDUAL TASK 1

Who gets reviewed first?

Machine learning for content moderation when you cannot read everything

A 4-minute executive briefing

THE PROBLEM

The constraint isn't detection. It's capacity.

~1%

of the daily stream is all a fixed review team can realistically read.

You can't review everything.

The volume beats any headcount you can hire.

You can't delete automatically.

Get it wrong and you have silenced ordinary people.

So the real question is triage.

Of everything that arrived today, what does a human see first?

THE SETUP

What I tested, and what I assumed

2 — **Two public collections of real comments**

About 25,000 posts from one platform and 160,000 from another, each already judged harmful or not by people.

2 — **Two automated sorting systems**

Built on different principles, both ranking content by how likely it is to break the rules.

5 — **Five separate tests of each**

Run on different slices of the data, so a good result can't be a fluke of one lucky sample.

Key assumption The machine's job is to rank, not to decide. A human still makes every call — so the cost of a wrong ranking is a wasted review, not an unjust ban.

FINDING 01

The obvious scoreboard lies

The 'better' system

94.5%

accurate overall

13%

of harmful posts caught

The simpler system

92.2%

accurate overall

41%

of harmful posts caught

Flag nothing at all and you still score 94%.

Harmful content is under 6% of the data.

Accuracy rewards a system for being quiet. We need one that is right when it speaks.

FINDING 02

The two systems were the same

Measured properly, the gap between them was small enough to be chance. They sort content into essentially the same order. The only thing that differs is where you draw the cut-off line — how aggressive you are willing to be about what reaches a reviewer.

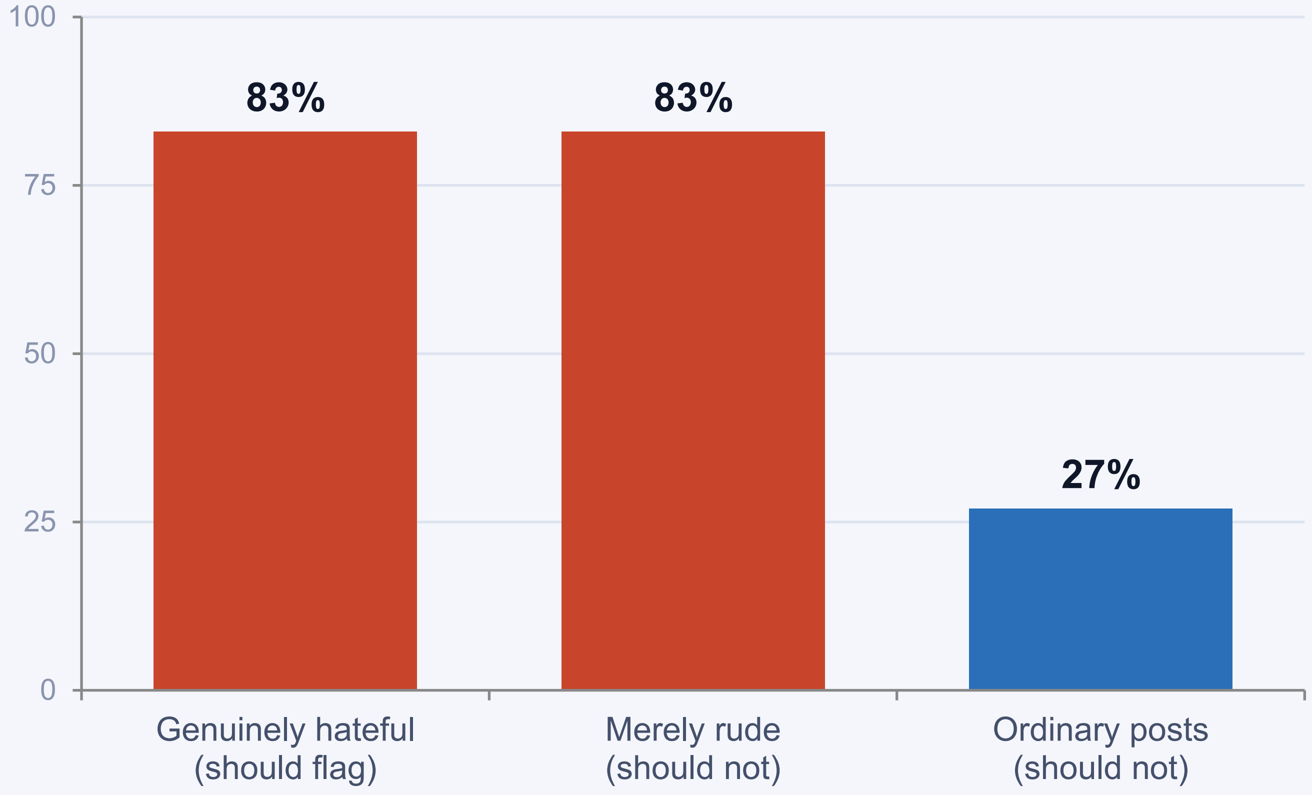
One ranks. The other ranks the same way. You are not choosing a model.

This isn't an engineering decision.

It's a staffing decision.

FINDING 03 · THE ONE THAT MATTERS

Then I moved it to a different platform



73%

of everything was flagged. The true rate of harmful content was under 6%.

It cannot tell them apart.

Genuinely hateful posts and merely rude ones were flagged at the identical rate — because the data it learned from never separated the two.

WHY IT MATTERS

In production, that is not a slightly worse score

Mass wrongful removals

Three quarters of ordinary posts pulled from a legitimate community.

An appeals backlog you cannot staff

Every wrongful removal becomes a complaint, and complaints need people too.

Regulatory and press attention

Over-blocking is the failure that makes the news, not under-blocking.

And it was invisible

Every test on the original dataset said the system was excellent.

RECOMMENDATION

What I would do

01

Ship the simpler system

It ranks just as well, trains for a fraction of the cost, and when someone appeals a removal you can actually explain why it was flagged.

02

Treat the cut-off as an operational dial

Tie it to how many reviewers you have this week. It is a capacity setting, not a fixed property of the model.

03

Never sign off on one source of labels

Validate every moderation model against data it has never seen from a platform it was not trained on.

WHAT I WOULD CAVEAT

Two limits on everything I just said

English text only

Both collections are written English. The platform I have in mind is short video, in dozens of languages. Nothing here transfers for free.

Fair on average is not fair

A system that looks even overall can still be far worse for one community. Before launch I would want the error rates broken down by dialect and speaker group.

Measure the queue, not the average.